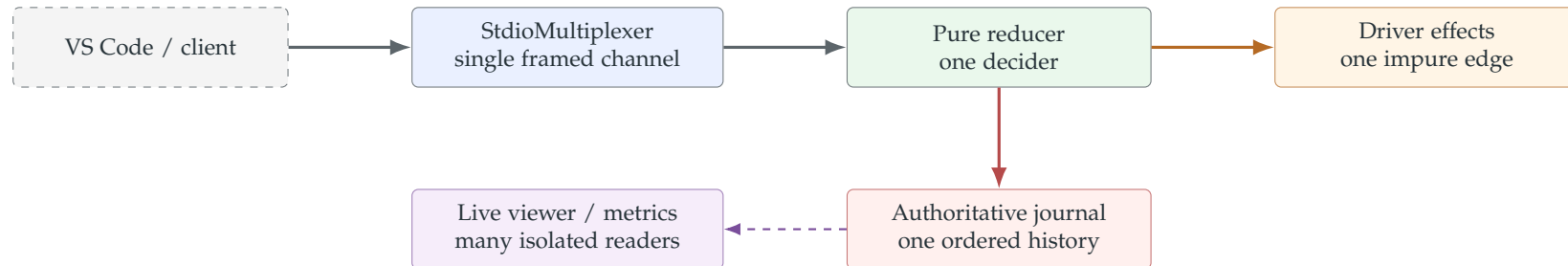


STS2 Design Visuals v2

Reviewed target architecture for a SQL Tools Service that explains itself while it runs



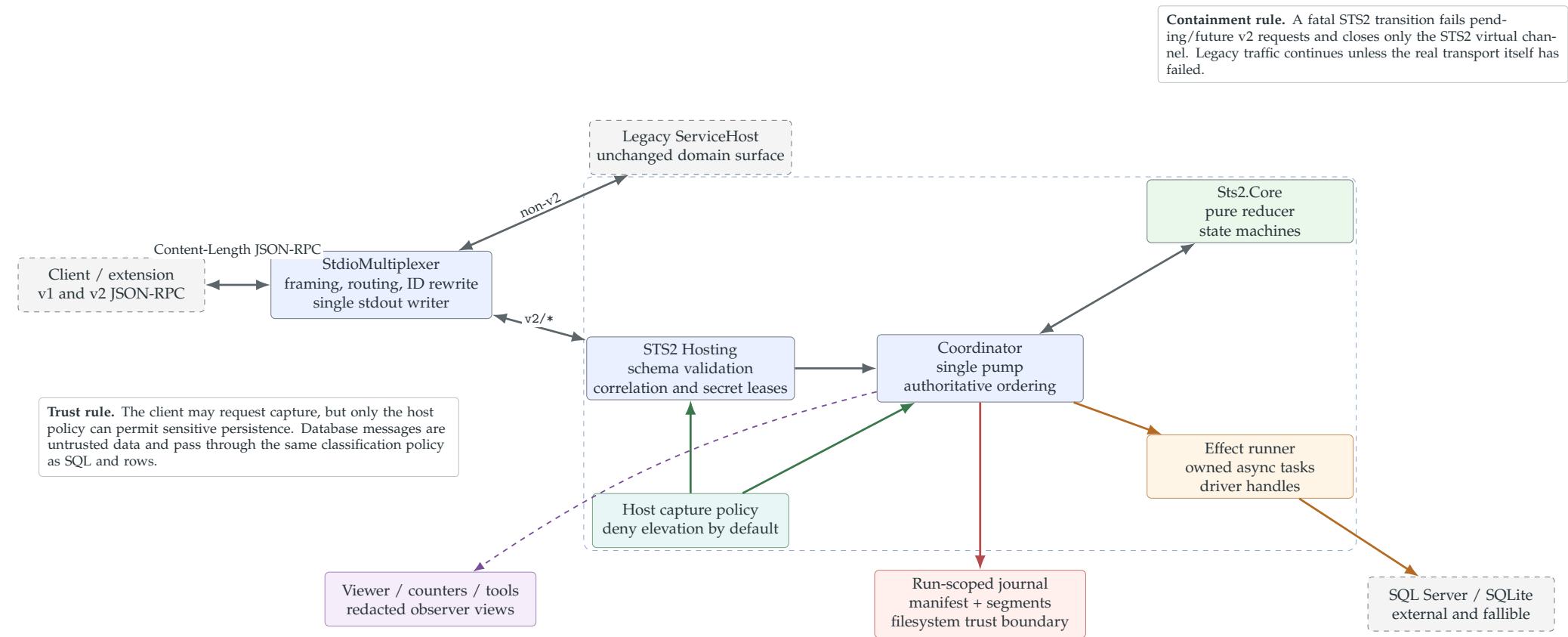
Design test: every important adjective has an executable meaning. *deterministic* means strict replay; *write-ahead* means a pump barrier; *bounded* means no driver advance without credit; *isolated* means a stuck observer cannot stall the pump; *private* means host policy controls persistence.

runtime pure Core journal effect edge observer policy/security

Companion files: 01_TECHNICAL_REVIEW.md, 03_TARGET_DESIGN.md, and 05_NEXT_STEPS.md.

1. System context and trust boundaries

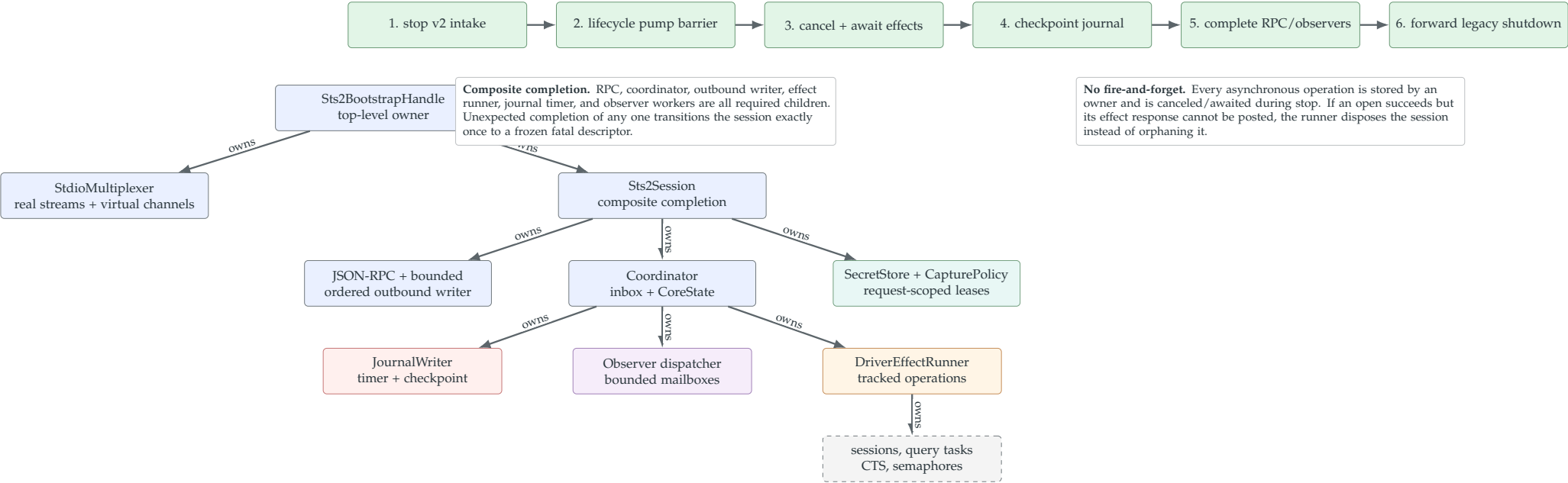
One process and one stdio channel, with explicit persistence and privacy boundaries.



2. Ownership and orderly shutdown

A single ownership tree prevents orphan tasks, sessions, secrets, and observer workers.

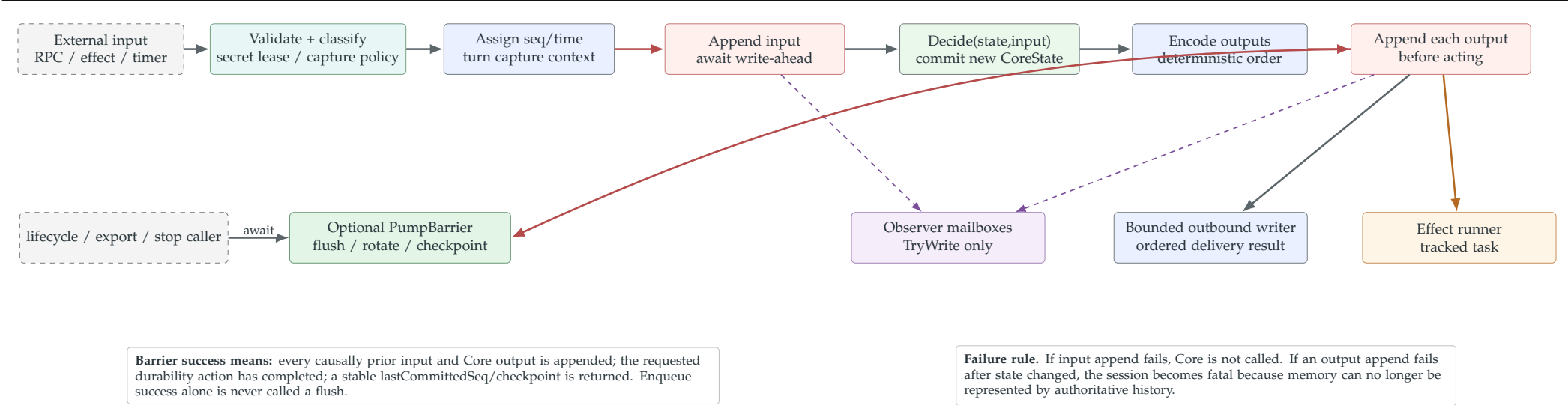
STS2 reviewed target design
branch snapshot c9fbd1e40ec8



3. One pump turn and the durability barrier

The journaled input and every Core output form one ordered commit group.

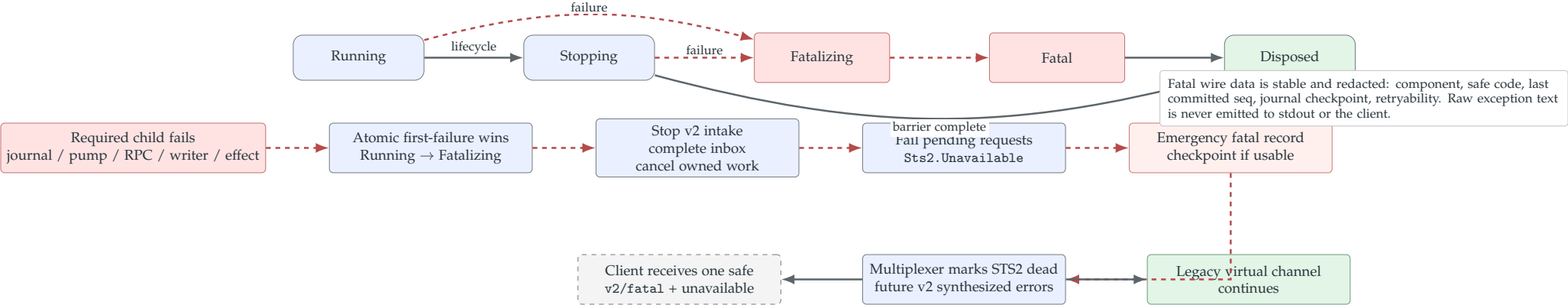
STS2 reviewed target design
branch snapshot c9fbd1e40ec8



4. Fatal containment and pending-request drain

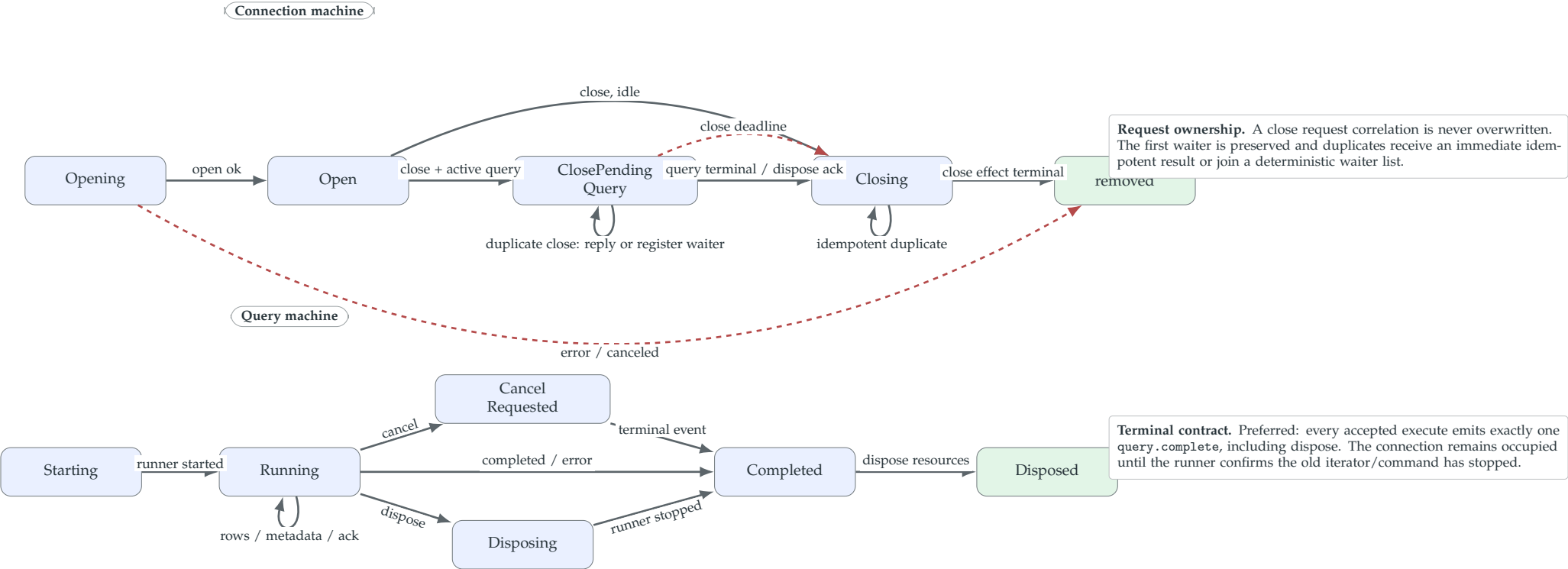
A failed STS2 becomes unavailable; it does not become a black hole and does not kill legacy.

STS2 reviewed target design
branch snapshot c9fbd1e40ec8



5. Connection and query lifecycle

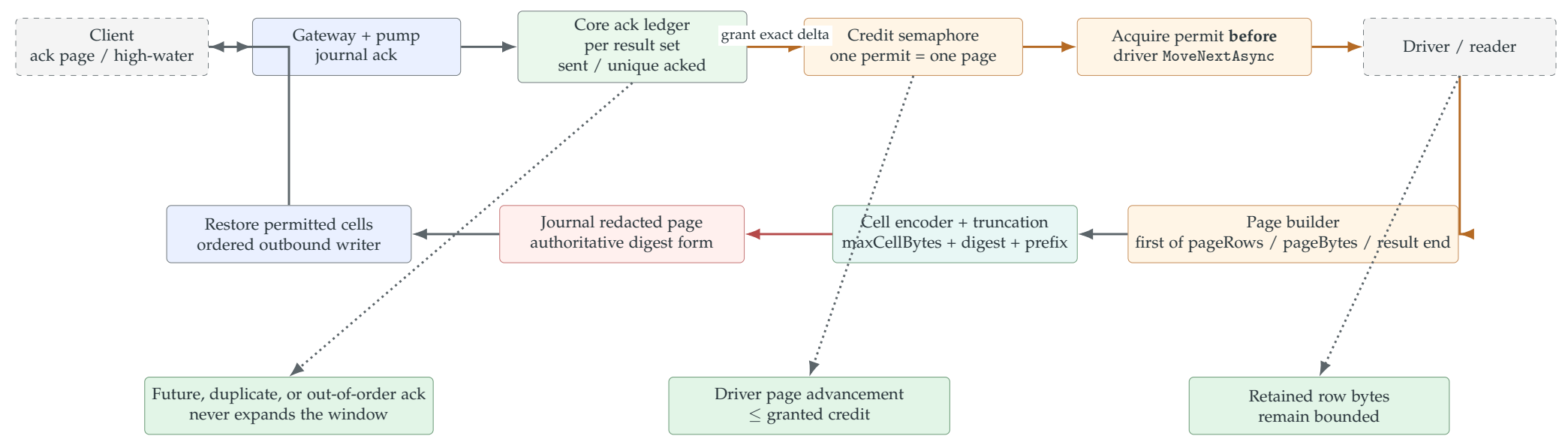
Recommended target states make close, dispose, timeout, and waiter ownership explicit.



6. Backpressure at the real driver edge

Credit is permission to request a page, not merely permission to send a notification.

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branch snapshot c9fbd1e40ec8



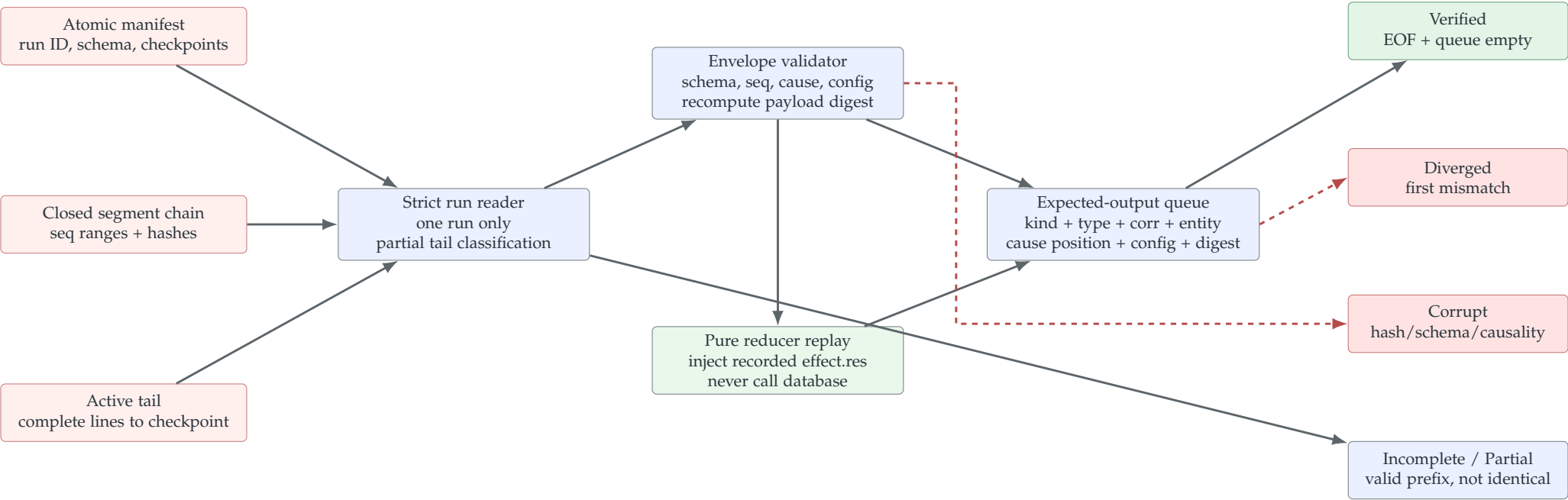
A high-water ack is clamped to the highest page actually sent for that result set. A per-page ack identifies its page and is idempotent. Credit is derived from unique sent minus unique acknowledged pages. The invariant is tested with an instrumented driver `MoveNext` counter, not only by counting outbound notifications.

7. Run-scoped journal and strict replay

Time travel may be partial; verification must prove a complete, internally valid run.

Exact-run rule. A manifest selects exactly one run directory. Readers, invariant checks, replay, and export never glob all `journal-*.jsonl` files from a shared folder.

Identity rule. Replay proves authoritative output envelopes. It does not claim identical runtime overlays, transport delivery, timestamps, database execution, or restored private values.



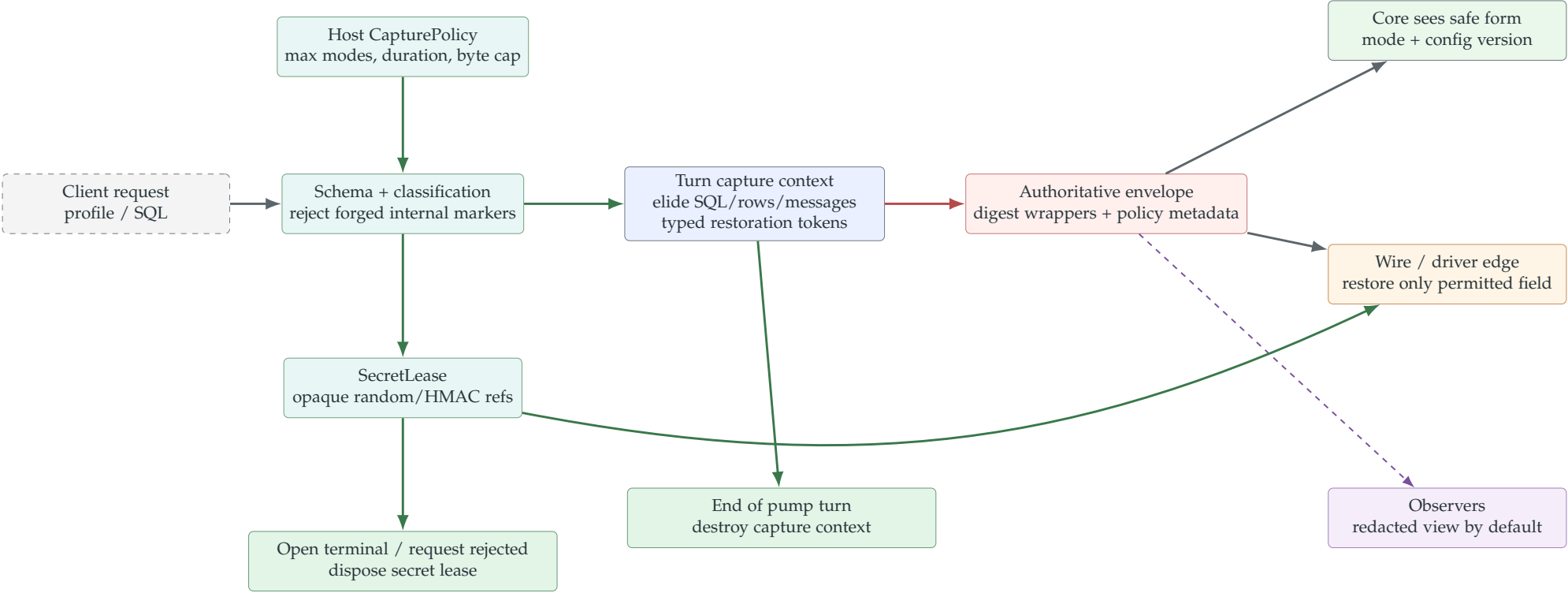
8. Privacy by policy and scoped restoration

Sensitive originals have short, owned lifetimes and never reach the authoritative journal by accident.

STS2 reviewed target design
branch snapshot c9fbd1e40ec8

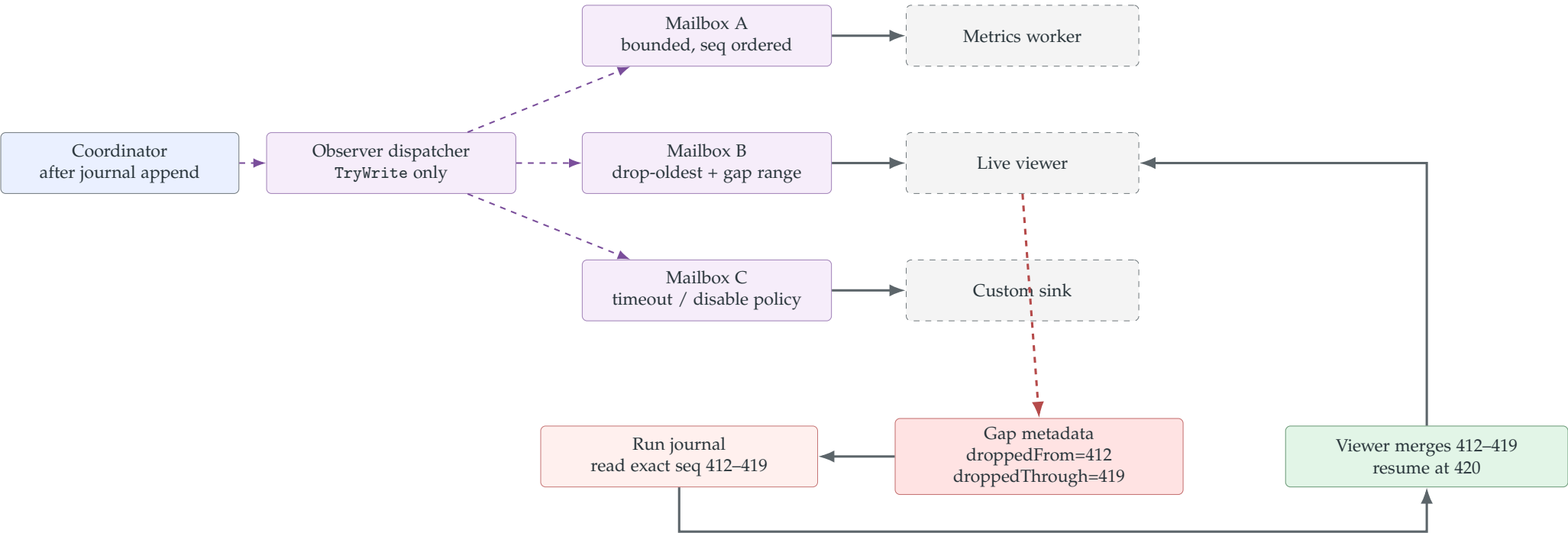
Product default is digest/digest and runtime elevation is denied. A client request cannot exceed host policy. Permitted elevation records reason, actor, version, bounds, and expiry.

Provider exceptions and server messages are classified data too. Stable safe codes/details enter the journal; governed raw diagnostics, when enabled, live in a separate protected channel and never in stdout.



9. Observer isolation and exact viewer resynchronization

The pump publishes references; observer code runs behind bounded mailboxes.



A sink that never completes cannot stall request processing. Its mailbox fills according to policy, faults/drops are counted, and the worker may be disabled. Session shutdown tracks and bounds every worker.

Observer registration declares the view: metadata-only, redacted authoritative, or a governed full view. Custom sinks do not automatically receive restored SQL, rows, or secrets.

10. Coherent export and verification

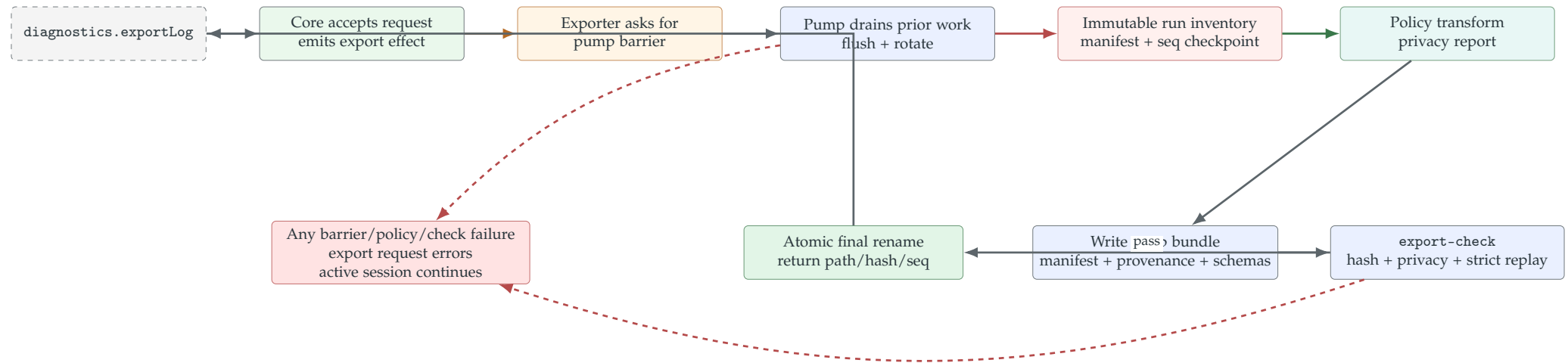
Export freezes an exact checkpoint, then copies and transforms outside the pump.

STS2 reviewed target design

branch snapshot c9fbd1e40ec8

The exporter never enumerates an active shared directory. It receives an exact file list for one run and one checkpoint. Full-capture source is transformed according to export policy; a `safeMode` label is never a substitute for transformation.

`export-check` validates semantic replay, not only self-consistent hashes. A bundle whose journal and manifest were both maliciously rewritten still fails the authoritative output comparison or privacy policy.



11. Release evidence and adoption ladder

The exact merge candidate, not mutable prose, earns the preview tag.

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branch snapshot c9fbd1e40ec8

CI evidence records exact SHA and merge base, SDK/dependency lock, SQL Server image digest, test counts, effective simulator seed count/range, mutation reports, perf environment, and generated artifact hashes.

A workflow on `pull_request.branches: sts2/main` does not protect a final PR whose base is `main`. The required gate must run on and be attached to the actual merge candidate.

